# Weak convergence and the moment method

Def: Let $\mathcal{P}$ be the space of (Borel) probability measures on $\mathbb{R}$.

$\mu_n \in \mathcal{P}$ converges <u>weakly</u> to $\mu \in \mathcal{P}$ as $n \to \infty$ if, for any continuous, bounded $f: \mathbb{R} \to \mathbb{R}$,

$$\int f \, d\mu_n \to \int f \, d\mu.$$

(1) This topology of weak convergence is <u>metrizable</u>, i.e. there is a metric $d(\cdot,\cdot)$ on $\mathcal{P}$ s.t. $\mu_n \to \mu$ weakly iff $d(\mu_n, \mu) \to 0$. One choice is Lévy distance:

$$d(\mu_n, \mu) = \inf\{\varepsilon > 0 : F_n(x-\varepsilon) - \varepsilon \le F(x) \le F_n(x+\varepsilon) + \varepsilon\}$$

where $F_n, F$ are the CDFs of $\mu_n, \mu$.

(2) Open sets of $(\mathcal{P}, d)$ generate a Borel $\sigma$-algebra on $\mathcal{P}$.

A <u>random measure</u> $\mu_n$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a Borel-measurable map $\Omega \mapsto \mathcal{P}$.

(We'll usually write $\mu_n \in \mathcal{P}$, similar to $X \in \mathbb{R}$ for a r.v. $X$.)

Then $\mu_n \to \mu$ weakly a.s. if $d(\mu_n, \mu) \to 0$ a.s. as $n \to \infty$.

Prop: The following two statements are equivalent:

(a) $\mu_n \to \mu$ weakly a.s.

(b) For each fixed continuous bounded $f: \mathbb{R} \to \mathbb{R}$,

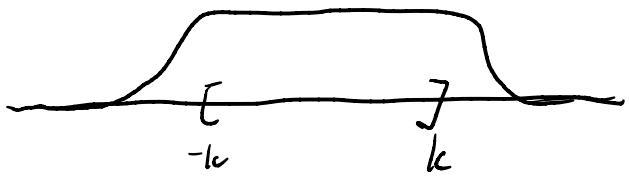
$$\int f \, d\mu_n \to \int f \, d\mu \quad \text{a.s.}$$

Proof: (a) $\Rightarrow$ (b) is immediate.

(b) $\Rightarrow$ (a): Let $C_0(\mathbb{R}) = \{ f: \mathbb{R} \to \mathbb{R} \text{ continuous, compactly supported} \}$.

$C_0(\mathbb{R})$ is separable under $\|\cdot\|_\infty$: Let $C$ be countable dense subset.

For each $k \in \mathbb{N}$, let $\phi_k \in C_0(\mathbb{R})$ be s.t. $1 \geq \phi_k \geq \mathbb{1}_{[-k,k]}$.




(b) $\Rightarrow \mathbb{P}\left[ \int g \, d\mu_n \to \int g \, d\mu \ \forall g \in C \cup \{\phi_k : k \in \mathbb{N}\} \right] = 1$.

On this event, for any $f \in C_0(\mathbb{R})$: Let $g \in C$ satisfy

$$\|g - f\|_\infty < \varepsilon \Rightarrow \left| \int f \, d\mu_n - \int f \, d\mu \right| \leq 2\varepsilon + \left| \int g \, d\mu_n - \int g \, d\mu \right|.$$

Take $n \to \infty$, $\varepsilon \to 0$: $\int f \, d\mu_n \to \int f \, d\mu \ \forall f \in C_0(\mathbb{R})$.

For any cont. bdd. $f: \mathbb{R} \to \mathbb{R}$: Let $k \in \mathbb{N}$ satisfy

$\mu([-k,k]) > 1 - \varepsilon$. Note $f\phi_k \in C_0(\mathbb{R})$.

$$\Rightarrow \left|\int f\,d\mu_n - \int f\,d\mu\right| \le \left|\int f\phi_k\,d\mu_n - \int f\phi_k\,d\mu\right| + \|f\|_\infty \cdot \left(\int (1-\phi_k)\,d\mu_n + \int (1-\phi_k)\,d\mu\right)$$

Take $n\to\infty$, $\varepsilon\to 0$: $\int f\,d\mu_n \to \int f\,d\mu$ $\forall$ cont. bdd. $f$. ∎

**Prop:** Suppose $\mu$ has compact support and $\int x^k\,d\mu_n \to \int x^k\,d\mu$ a.s. for each $k\ge 1$. Then $\mu_n \to \mu$ weakly a.s.

**Proof:** Suppose $\mu([-M,M]) = 1$. The given condition implies

$$\mathbb{P}\left[\int x^k\,d\mu_n \to \int x^k\,d\mu \ \ \forall k\ge 1\right] = 1.$$ On this event,

$$\int x^{2j}\,\mathbb{1}\{|x|\ge M+1\}\,d\mu_n \le \int \frac{x^{2j+2k}}{(M+1)^{2k}}\,d\mu_n \longrightarrow \frac{\int x^{2j+2k}\,d\mu}{(M+1)^{2k}} \le \frac{M^{2k}}{(M+1)^{2k}}\cdot M^{2j}.$$

Take $k\to\infty$: $\int x^{2j}\,\mathbb{1}\{|x|\ge M+1\}\,d\mu_n \to 0$ $\forall j\ge 1$.

For any cont. bdd. $f:\mathbb{R}\to\mathbb{R}$:

By Weierstrass approx., $\exists$ polynomial $p:\mathbb{R}\to\mathbb{R}$ s.t. $\sup_{|x|\le M+1} |f(x)-p(x)| < \varepsilon$.

$$\Rightarrow \left|\int f\,d\mu_n - \int f\,d\mu\right| \le \left|\int p\,d\mu_n - \int p\,d\mu\right| + \int |f-p|\,d\mu_n + \int |f-p|\,d\mu$$

$$\le \left|\int p\,d\mu_n - \int p\,d\mu\right| + 2\varepsilon + \int |f(x)-p(x)|\cdot \mathbb{1}\{|x|\ge M+1\}\,d\mu_n$$

Take $n\to\infty$, $\varepsilon\to 0$: $\int f\,d\mu_n \to \int f\,d\mu$ $\forall$ cont. bdd. $f$. ∎

Recall: If $X \in \mathbb{R}^{n \times n}$ is Wigner w/ ESD $\mu_n$, then

$$\mathbb{E} \int x^k \, d\mu_n \to \int x^k \, d\mu_{sc} \quad \text{(Lecture 1)}$$

To complete Wigner's Thm, need to strengthen this to a.s. convergence.

Lemma: For each $k \geq 1$, there exists a constant $C_k > 0$ s.t.

$$\mathrm{Var}\left[\int x^k \, d\mu_n\right] \leq \frac{C_k}{n^2}.$$

Proof: Write $\int x^k \, d\mu_n = \frac{1}{n} \mathrm{Tr}\, X^k = \frac{1}{n} \sum_{i_1, \dots, i_k = 1}^{n} X_{i_1 i_2} \cdots X_{i_k i_1}$.

$$\mathrm{Var}\left[\int x^k \, d\mu_n\right] = \mathbb{E}\left(\frac{1}{n} \mathrm{Tr}\, X^k\right)^2 - \left(\mathbb{E} \frac{1}{n} \mathrm{Tr}\, X^k\right)^2$$

$$= \frac{1}{n^2} \sum_{\substack{i_1, \dots, i_k \\ j_1, \dots, j_k}} \Big( \underbrace{\mathbb{E}\, X_{i_1 i_2} \cdots X_{i_k i_1} X_{j_1 j_2} \cdots X_{j_k j_1} - \mathbb{E}\, X_{i_1 i_2} \cdots X_{i_k i_1} \cdot \mathbb{E}\, X_{j_1 j_2} \cdots X_{j_k j_1}}_{(*)} \Big)$$

Summands are indexed by pairs of walks $(i_1, \dots, i_k)$ and $(j_1, \dots, j_k)$ on the complete graph on $\{1, \dots, n\}$.

- If walks $(i_1, \dots, i_k)$ and $(j_1, \dots, j_k)$ are edge-disjoint then $X_{i_1 i_2} \cdots X_{i_k i_1}$ is independent of $X_{j_1 j_2} \cdots X_{j_k j_1} \Rightarrow (*) = 0$.
- If a distinct edge $e$ is traversed only once across both walks, then $\mathbb{E} X_e = 0$, hence $(*) = 0$

- The remaining walk-pairs have # distinct edges $\le k$ and form a connected graph, so # distinct vertices $\le k+1$.

  If # distinct vertices $= k+1$, then the graph is a tree w/ each edge traversed twice.

  $(i_1, \dots, i_k)$ and $(j_1, \dots, j_k)$ are not edge-disjoint

  $\Rightarrow$ some edge is traversed once by each walk, but this is impossible since the walks return back to $i_1, j_1$.

  Thus: # distinct vertices $\le k$.

The number of such walk-pairs is $\le C_k \cdot n^k$

For each such walk pair: Since $X_{ij} = \frac{Z_{ij}}{\sqrt{n}}$ where $Z_{ij}$ has bounded moments, $(*) \le \frac{C_k'}{n^k}$

So $\mathrm{Var}\left[\int x^k d\mu_n\right] \le \frac{1}{n^2} \times C_k n^k \times \frac{C_k'}{n^k} \le \frac{C_k C_k'}{n^2}$. ∎

Proof of Wigner's Thm: For each $k \ge 1$:

$$\mathbb{P}\left[\left|\int x^k d\mu_n - \mathbb{E}\int x^k d\mu_n\right| \ge \varepsilon\right] \le \frac{\mathrm{Var}\left[\int x^k d\mu_n\right]}{\varepsilon^2} \le \frac{C_k}{\varepsilon^2} \cdot \frac{1}{n^2}.$$

and $\mathbb{E}\int x^k d\mu_n \to \int x^k d\mu$. Then by Borel-Cantelli Lemma,

$$\left|\int x^k d\mu_n - \int x^k d\mu\right| \le \left|\int x^k d\mu_n - \mathbb{E}\int x^k d\mu_n\right| + \left|\mathbb{E}\int x^k d\mu_n - \int x^k d\mu\right|$$

$< 2\varepsilon$ a.s. for all large $n$.

Taking $\varepsilon \to 0$: $\int x^k \, d\mu_n \to \int x^k \, d\mu$ a.s. $\forall k \geq 1$

So $\mu_n \to \mu$ weakly a.s. ∎

Note: If eigenvalues $\lambda_1, \ldots, \lambda_n$ were independent r.v.'s, then we would have $\mathrm{Var}\left[\int x^k \, d\mu_n\right] = \mathrm{Var}\left[\frac{1}{n}\sum_{i=1}^n \lambda_i^k\right] \asymp \frac{1}{n}$.

We have shown instead: $\mathrm{Var}\left[\int x^k \, d\mu_n\right] \leq \frac{1}{n^2}$

The eigenvalues $\lambda_1, \ldots, \lambda_n$ are more rigid than i.i.d r.v.'s, and $\frac{1}{n}\sum_{i=1}^n \lambda_i^k$ has stronger concentration around its mean.

## Concentration of spectral measure

For a large class of Wigner matrices, concentration of $\frac{1}{n}\sum_{i=1}^n f(\lambda_i)$ and its mean can be understood via general principles of concentration-of-measure.

Def: A distribution $\nu$ satisfies dimension-free Lipschitz concentration if there exist constants $C, c > 0$ s.t.:

For any $m \geq 1$ and $f : \mathbb{R}^m \to \mathbb{R}$ satisfying

$$|f(z) - f(z')| \leq L \cdot \|z - z'\|_2 \quad \forall z, z' \in \mathbb{R}^m$$

(i.e. $f$ is L-Lipschitz), we have

$$\mathbb{P}\left[\left|f(z_1,\dots,z_m) - \mathbb{E}f(z_1,\dots,z_m)\right| \geq t\right] \leq C e^{-\frac{ct^2}{L^2}}$$

when $z_1,\dots,z_m \overset{iid}{\sim} \nu$.

Thm: $\mathcal{N}(0,1)$ satisfies dimension-free Lipschitz concentration.

Proof: Let $Z = (z_1,\dots,z_m)$, $z_1,\dots,z_m \overset{iid}{\sim} \mathcal{N}(0,1)$.

(i) Suppose $f: \mathbb{R}^m \to \mathbb{R}$ L-Lipschitz, continuously-differentiable.

Let $Z'$ be independent copy of $Z$. By Jensen's ineq., $\forall \lambda \geq 0$,

$$\mathbb{E}\, e^{\lambda(f(Z) - \mathbb{E}f(Z))} = \mathbb{E}_Z e^{\lambda(f(Z) - \mathbb{E}_{Z'} f(Z'))}$$

$$\leq \mathbb{E}_{Z,Z'}\, e^{\lambda(f(Z) - f(Z'))}$$

Write $f(Z) - f(Z') = \int_0^{\pi/2} \frac{d}{d\theta} f(\underbrace{Z'\cos\theta + Z\sin\theta}_{:= Z_\theta})\, d\theta$

By Jensen's ineq.,

$$e^{\lambda(f(Z) - f(Z'))} = e^{\frac{\pi\lambda}{2} \cdot \int_0^{\pi/2} \frac{d}{d\theta} f(Z_\theta) \cdot \frac{2}{\pi} d\theta}$$

$$\leq \int_0^{\pi/2} e^{\frac{\pi\lambda}{2} \cdot \frac{d}{d\theta} f(Z_\theta) \cdot \frac{2}{\pi}} d\theta$$

$$= \frac{2}{\pi} \int_0^{\pi/2} e^{\frac{\pi\lambda}{2} \nabla f(Z_\theta)^T (\underbrace{-Z'\sin\theta + Z\cos\theta}_{:= Z'_\theta})}\, d\theta$$

Note: $\mathbb{E}\, z_\theta z_\theta'^{\top} = \sin\theta\cos\theta\,(-\mathbb{E}\, z' z'^{\top} + \mathbb{E}\, z z^{\top}) = 0$

$\mathbb{E}\, z_\theta z_\theta^{\top} = \mathbb{E}\, z_\theta' z_\theta'^{\top} = I$

$\Rightarrow z_\theta, z_\theta' \in \mathbb{R}^m$ are independent $\mathcal{N}(0, I)$ vectors.

$$\mathbb{E}\, e^{\lambda(f(z) - \mathbb{E}f(z))} \le \frac{2}{\pi}\int_0^{\pi/2} \mathbb{E}\, e^{\frac{\pi\lambda}{2}\nabla f(z_\theta)^{\top} z_\theta'}\, d\theta$$

$$\le \frac{2}{\pi}\int_0^{\pi/2} \mathbb{E}\, e^{\frac{\pi^2\lambda^2}{8}\|\nabla f(z_\theta)\|_2^2}\, d\theta$$

$$\le e^{\frac{\pi^2 L^2}{8}\cdot\lambda^2}$$

$$\Rightarrow \mathbb{P}[f(z) - \mathbb{E}f(z) \ge t] \le e^{-\lambda t}\, \mathbb{E}\, e^{\lambda(f(z) - \mathbb{E}f(z))}$$

$$\le e^{-\lambda t + \frac{\pi^2}{8}L^2\cdot\lambda^2}$$

$$= e^{-\frac{ct^2}{L^2}} \quad (\text{pick } \lambda \asymp \tfrac{t}{L^2})$$

Same argument holds for $-f$: $\mathbb{P}[f(z) - \mathbb{E}f(z) \le t] \le e^{-\frac{ct^2}{L^2}}$.

(iii) For general $f: \mathbb{R}^m \to \mathbb{R}$ $L$-Lipschitz:

Let $f_\varepsilon(x) = \mathbb{E}_{z\sim\mathcal{N}(0,I)} f(x + \varepsilon z) = \int f(y)\left(\frac{1}{\sqrt{2\pi\varepsilon^2}}\right)^m e^{-\frac{\|y-x\|_2^2}{2\varepsilon^2}}\, dy$.

- $f_\varepsilon(x)$ is continuously differentiable
- $|f_\varepsilon(x) - f_\varepsilon(x')| \le \mathbb{E}_z\, |f(x + \varepsilon z) - f(x' + \varepsilon z)| \le L\cdot\|x - x'\|_2$
- $|f_\varepsilon(x) - f(x)| \le \mathbb{E}_z\, |f(x + \varepsilon z) - f(x)| \le L\cdot\mathbb{E}_z\|\varepsilon z\|_2$

$\le L\sqrt{m}\cdot\varepsilon$ uniform in $x$.

$$\Rightarrow \mathbb{P}\left[|f(z) - \mathbb{E}f(z)| \geq t\right]$$

$$\leq \mathbb{P}\left[|f_\varepsilon(z) - \mathbb{E}f_\varepsilon(z)| \geq t - 2L\sqrt{n}\varepsilon\right] \leq 2e^{-\frac{c(t-2L\sqrt{n}\varepsilon)^2}{L^2}}$$

by (i). Take $\varepsilon \to 0$. ∎

Remark: More generally, $\nu$ satisfies dimension-free Lipschitz concentration if it has Lebesgue density $e^{-U(x)+V(x)}$ where $U''(x) \geq c > 0$ $\forall x \in \mathbb{R}$, and $V: \mathbb{R} \to \mathbb{R}$ is Lipschitz. (In this case $\nu$ satisfies a log-Sobolev inequality.)

To apply this to eigenvalues of a Wigner matrix:

Lemma (Hoffman-Wielandt): Suppose $X, X' \in \mathbb{R}^{n \times n}$ w/ eigenvalues $\lambda_1 \leq \dots \leq \lambda_n$ and $\lambda_1' \leq \dots \leq \lambda_n'$. Then

$$\sum_{i=1}^n (\lambda_i - \lambda_i')^2 \leq \mathrm{Tr}\,(X - X')^2.$$

Proof: Let $X = \sum_{i=1}^n \lambda_i v_i v_i^T$, $X' = \sum_{j=1}^n \lambda_j' v_j' v_j'^T$. Then

$$\mathrm{Tr}\, XX' = \sum_{i,j=1}^n \lambda_i \lambda_j' \underbrace{|v_i^T v_j'|^2}_{:= p_{ij}}.$$

$P = (p_{ij})_{i,j=1}^n$ is a doubly-stochastic matrix (all rows/columns sum to 1) $\Rightarrow$ $P$ is a convex combination of permutations $\Pi$.

For each fixed $\pi$, by rearrangement inequality,

$$\sum_{i=1}^n \lambda_i \lambda'_{\pi(i)} \pi_{i,\pi(i)} \le \sum_{i=1}^n \lambda_i \lambda'_i$$

$$\Rightarrow \operatorname{Tr} XX' \le \sum_{i=1}^n \lambda_i \lambda'_i$$

$$\Rightarrow \operatorname{Tr}(X-X')^2 = \operatorname{Tr} X^2 + \operatorname{Tr} X'^2 - 2\operatorname{Tr} XX'$$

$$\ge \sum_{i=1}^n \lambda_i^2 + \lambda_i'^2 - 2\lambda_i \lambda'_i = \sum_{i=1}^n (\lambda_i - \lambda'_i)^2. \quad \blacksquare$$

Cor: Suppose $X$ is Wigner, $X_{ij} = \frac{z_{ij}}{\sqrt{n}}$ where

$\left\{\frac{z_{ii}}{\sqrt{2}}\right\}_{i=1}^n, \{z_{ij}\}_{i<j} \overset{iid}{\sim} \nu$ satisfy dim-free Lip.-concentration.

Then $\exists$ constants $C, c > 0$ s.t. $\forall$ 1-Lipschitz $f: \mathbb{R} \to \mathbb{R}$,

$$\mathbb{P}\left[\left|\frac{1}{n}\sum_{i=1}^n f(\lambda_i) - \mathbb{E}\frac{1}{n}\sum_{i=1}^n f(\lambda_i)\right| \ge t\right] \le Ce^{-\frac{ct^2n^2}{L^2}}.$$

[Thus $\frac{1}{n}\sum_{i=1}^n f(\lambda_i) = O_{\mathbb{P}}\left(\frac{L}{n}\right)$.]

Proof: Define $F\left(\underbrace{\left\{\frac{z_{ii}}{\sqrt{2}}\right\}_{i=1}^n, \{z_{ij}\}_{i<j}}_{:= z \in \mathbb{R}^{\frac{n(n+1)}{2}} \text{ as a vector}}\right) = \frac{1}{n}\sum_{i=1}^n f(\lambda_i)$.

By Hoffman-Wielandt ineq.,

$$\left|\frac{1}{n}\sum_{i=1}^n f(\lambda_i) - \frac{1}{n}\sum_{i=1}^n f(\lambda'_i)\right|$$

$$\leq \frac{L}{n} \sum_{i=1}^{n} |\lambda_i - \lambda_i'| \leq L \left( \frac{1}{n} \sum_{i=1}^{n} |\lambda_i - \lambda_i'|^2 \right)^{1/2}$$

$$\leq L \cdot \left( \frac{1}{n} \operatorname{Tr} (X - X')^2 \right)^{1/2}$$

$$= L \cdot \left( \frac{1}{n} \sum_{i,j=1}^{n} \left( \frac{z_{ij}}{\sqrt{n}} - \frac{z_{ij}'}{\sqrt{n}} \right)^2 \right)^{1/2} = \frac{\sqrt{2} \cdot L}{n} \cdot \|z - z'\|_2.$$

Thus $F : \mathbb{R}^{\frac{n(n+1)}{2}} \to \mathbb{R}$ is $\frac{\sqrt{2} \cdot L}{n}$-Lipschitz.

Result follows from Lipschitz concentration. ∎

**Problem 2**: Let $X \in \mathbb{R}^{n \times n}$ be a Wigner matrix, w/ diagonal $x_{ii} = 0 \quad \forall i = 1, \ldots, n$.

(a) Show that $\operatorname{Tr} X^2 - \mathbb{E} \operatorname{Tr} X^2$ (w/o $\frac{1}{n}$ normalization) converges to a normal limit, as $n \to \infty$, and compute its variance.

(b) For each $k \geq 1$, show that $\lim_{n \to \infty} \mathbb{E} (\operatorname{Tr} X^3)^k$ exists, and derive an explicit form for this limit. Using this, show that $\operatorname{Tr} X^3$ converges in distribution to a normal limit, as $n \to \infty$, and compute its variance.